INTRODUCTION

Many "As Manufactured" drawings supplied with Gas Turbine equipment require field changes or revisions during the installation and start-up phases, or during Field Modification Instruction (FMI) work completion. It is essential that these changes be documented in accordance with this instruction (EM 5185) so that all necessary revisions can be accurately identified, approved and incorporated into the revised drawings.

Until recently, the practice of modifying "As Manufactured" drawings involved marking prints using only Black Ink. In an effort to improve the readability of marked up prints, all changes are now required to be made using a Color Coding system. Color Coding Instructions are detailed on page 3 of this instruction.

HEALTH AND SAFETY REQUIREMENTS

In all cases, the Health, Safety and Environmental Manual (HSEM) will be the document followed on the job site during installation, maintenance and repair of equipment.

Safety, health, and environmental protection must be an integral part of all department business operations, including design, construction, start-up and maintenance. Every reasonable effort will be made to create and maintain a safe and healthful work environment at each customer site.

GENERAL INFORMATION

"As Installed" drawings are "As Manufactured" drawings that have been marked up in the field to reflect the actual "As Installed" hardware configuration. Upon completion, they are to be returned to Installation and Field Services (I&FS) for review and subsequent reissue and distribution as revised "As Installed" drawings to correctly reflect the "As Installed" configuration. These drawings normally fall into one of two general categories (defined as Category I and Category II).

"AS INSTALLED" DRAWING CATEGORIES – DEFINITIONS

Category I

Drawings furnished as part of the Field Installation documentation package, which are returned from the field as marked up "As Installed" drawings, and which reflect (via the mark ups) the actual "As Installed" configuration of the unit or station. These drawings are then returned to the I&FS Installation Engineer.

Category II

Drawings furnished as part of a Field Modification Instruction (FMI) or Engineering Change Notice (ECN), and issued after the Commercial Operation Date (COD). These drawings are to be marked up and returned from the field after completion of a modification, uprating or conversion. These drawings are returned to the I&FS Customer Service Engineer.

PROCEDURE FOR MARKING UP "AS MANUFACTURED" DRAWINGS

CATEGORY I

Revisions are to be made on two (2) sets of the appropriate drawings using the Color Coding method described in this instruction.

Instructions

1. Detail all changes made in the field up to and including the commercial operating date, including all FMI's applied during the installation.

2. Furnish one (1) set of the "As Installed" marked up drawings to the designated Installation Engineer in Schenectady, NY, **No Later than Thirty (30) Days After the Date of Synchronization/Load.** In cases where a site is demobilized prior to completion, one (1) copy of all marked up drawings in existence at the time of demobilization are to be returned to I&FS. This is the responsibility of the site Lead/Startup Engineer, and is to be completed before leaving the site.

3. In addition to the marked up drawings, the Lead/Startup Engineer must also provide:

    a. Notation on all controls changes stating the authorization for the change (i.e., FRN, FMI, ENC, etc.).

    b. A list of the drawing numbers, identifying the section, sheet, and revision level of all marked up drawings.

    c. The "Completion Notice Card" listing all FMI's and ECN's completed before the unit went into commercial operation. This certifies that all other drawings remain as issued.

A second set of marked-up "As Installed" drawings is to be sent to the customer for use pending receipt of revised documentation from I&FS. The date of delivery, name and position of the individual to whom the drawings are sent is to be noted in the Start-up Report.

Note: A third set of Category I "As Installed" marked up drawings may be required for the local I&FS office if requested by management.

CATEGORY II

The cognizant Lead/Startup Engineer must provide two (2) sets of Category II field marked up "As Installed" drawings. One set is to be forwarded to the customer; and one set is to be forwarded to the Customer Service Engineer.

Note: A third set of Category II "As Installed" marked up drawings may be required if requested by the I&FS district office.

COLOR CODING

Follow the Color Coding instructions below when marking up drawings.

1. REMOVING items or lines – use only RED ink.
2. ADDING items or lines – use only BLACK ink OVER YELLOW background.
3. DO NOT USE any other color when marking up drawings.

NOTE: Major considerations in marking up drawings are that **changes must be:**

- Legible
- Understandable
- Complete